DEPLOYMENT MODULE

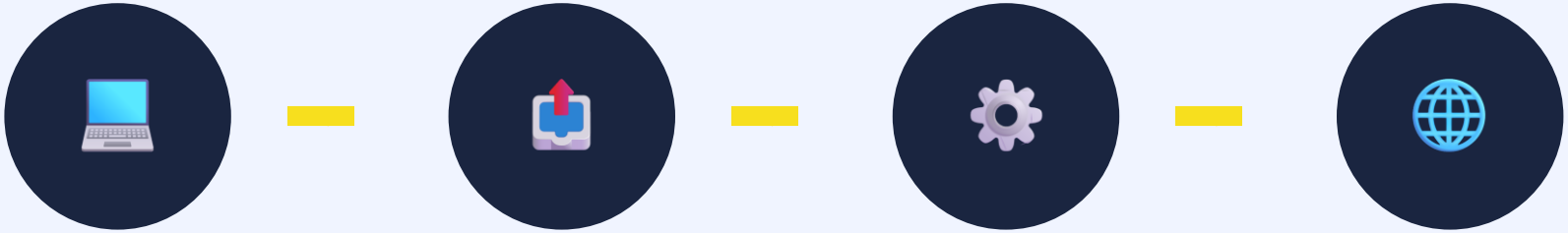
Week 9 · Fri

Deploying Your Website



Making your site public on the internet — for free

What is Deployment?



Your Laptop

Code works locally
(localhost only)

Push to GitHub

Code is stored
in the cloud

GitHub Pages Builds Site

Converts your repo
into a website

Live on Internet!

Anyone can visit
your URL

Deployment Options (from beginner to advanced)

★ GitHub Pages — free, static sites, perfect for this course

Netlify / Vercel — free tiers, easy CI/CD, supports more frameworks

Cloud servers (AWS, GCP, Azure) — full control, paid, advanced

GitHub Pages Overview

▶ 100% Free

For public repos

▶ No server needed

GitHub hosts for you

▶ Auto-updates

Push = site updates

▶ Custom domain

optional, free

▶ HTTPS included

Secure by default

Your GitHub Pages URL

```
# User/organization site
username.github.io


# Project site (most common)
username.github.io/repo-name


# Example for this class:
juma2024.github.io/book-search-app


# After enabling GitHub Pages:
Your site is live in ~2 minutes
Updates on every git push
```

Enable GitHub Pages — Step by Step

Settings → Pages → Deploy

1

Open your repo on GitHub

Go to github.com → your repo page (the Book Search App repo from Monday)

2

Click Settings tab

Top navigation: Code, Issues, Pull requests ... Settings. Click Settings.

3

Scroll to Pages section

Left sidebar: find "Pages" under Code and Automation section

4

Set Source to "main" branch

Under "Build and deployment" → Source: Deploy from branch → Branch: main → /(root)

5

Click Save

GitHub starts building. Wait ~2 minutes. Refresh the page to see your live URL.

Your GitHub Pages URL

juma2024.github.io/**book-search-app**

Your GitHub username	GitHub Pages domain	Your repository name (exact match)

⚠️ Repo name must be lowercase and use hyphens — spaces are not allowed

✅ The EXACT repo name becomes part of your URL — rename carefully

📁 index.html must be in the ROOT folder of your repo (not in /src or /public)

Static vs Dynamic — What Works?

✅ Works on GitHub Pages	✕ Does NOT Work
✓ HTML files (.html)	✕ PHP (server-side language)
✓ CSS stylesheets (.css)	✕ Python / Flask / Django
✓ JavaScript files (.js)	✕ Node.js / Express
✓ Images (PNG, JPG, SVG)	✕ Databases (MySQL, MongoDB)
✓ JSON data files	✕ File system writes
✓ Web fonts	✕ User authentication (sessions)
✓ Client-side frameworks (React, Vue with build step)	✕ .htaccess server config
✓ Fetch requests to external APIs	

index.html is Your Entry Point

▶ Must be named

"index.html"

exactly

▶ Must be in

the ROOT folder

of your repo

▶ Other pages

link relatively:

./about.html

Correct repo structure

```
book-search-app/
  index.html          <- ENTRY POINT
  style.css
  app.js
  README.md


# Wrong! (causes 404)
book-search-app/
  src/
    index.html        <- NOT found!


# Fix: move index.html to root OR
# set Pages source to /src folder
```

GitHub Actions — Automation & CI/CD



What is it?

Automated workflows that run when something happens in your repo (push, PR, schedule).



CI/CD Concept

CI = run tests automatically on every push. CD = deploy automatically when tests pass.



Workflow File

A YAML file in .github/workflows/ tells GitHub WHAT to run and WHEN to run it.

```yaml
on: push    # trigger: on every push
jobs:
  deploy:
    runs-on: ubuntu-latest
```

Lab Activity

Deploy Your Book Search App to GitHub Pages

1 Verify repo structure

index.html must be in root. Check with: ls -la

2 Push latest code

git add .
git commit -m "Final for deployment"
git push origin main

3 Enable GitHub Pages

GitHub repo → Settings → Pages → Source: main branch → Save

4 Wait for build

~2 minutes. Refresh Settings → Pages until you see the green URL.

5 Test your live site

Open the URL in a browser. Test search. Open on your phone!

6 Share your URL

Post your live site URL in the week9-deploy Cybercampus.

Common Issues & Fixes

Problem: 404 Not Found error

Fix: Check index.html is in the repo root. Rename it from Index.html or INDEX.HTML — casing matters!

Problem: Site shows old code after push

Fix: Hard refresh: Ctrl+Shift+R (Windows) or Cmd+Shift+R (Mac). GitHub Pages cache can lag 1-2 minutes.

Problem: GitHub Pages option is grey / not available

Fix: Repository must be PUBLIC. Go to Settings → General → Danger Zone → Change visibility → Public.

Problem: Images/CSS/JS not loading

Fix: All file paths must be relative (./style.css not /style.css). Absolute paths break on GitHub Pages.

Problem: API fetch returns error on live site

Fix: Check browser console. The API must allow CORS. Open Library API used in class supports CORS.

Week 9 Summary & Next Steps

This week you learned:

- Git basics: init, add, commit, log
- Branching and merging
- Working with GitHub remote
- GitHub Pages deployment
- Static vs dynamic site hosting
- GitHub Actions (concept)

Coming up:

Week 10

Team Project Kickoff — plan, repo setup, first commits

Week 11

Advanced JS — Promises, async/await, APIs in depth

Week 12

Backend intro — Node.js + Express server deployment

HOMEWORK: Add form validation to your team project contact/signup form (from Friday's Monday class notes). Share your GitHub Pages URL + PR link on Cybercampus by Sunday.